Unit-3 Supervised Machine Learning

*** 1-Marks Questions**

1. Define Machine Learning.
2. What is Supervised Learning?
3. What is Overfitting and Underfitting?
4. What is Bias and Variance?
5. What are Support Vectors in SVM?
6. What is meant by 'Bagging' in Random Forest?

*** 3-Marks Questions**

7. Differentiate between Classification and Regression with suitable examples.
8. Describe the Bias-Variance?
9. Explain Logistic Regression and describe how the Sigmoid function is used for classification.
10. Explain Naive Bayes algorithm
11. Write a program to Predict student result based on study hours using LightGBM.
12. Explain Linear Regression.

*** 5-Marks Questions**

13. Describe Decision Tree and Random Forest algorithms in detail, covering their working, splitting criteria, advantages and weaknesses.
14. Write a program to create end to end system.
15. Explain hyperparameter tuning in detail. Describe Grid Search, Random Search, Bayesian Optimization and K-Fold Cross-Validation with their pros and cons.
16. Write a Program to predict student result based on attendance in XGBoost.
17. Compare Scikit-learn, XGBoost and LightGBM in detail.
18. Write a program to perform Predict Sales based on Budget using Hyperparameter.

Unit-4: Deep Learning Implementation

*** 1-Marks Questions**

19. What is an Activation Function? Name any two.
20. What is Deep Learning?
21. Define Neural Network.
22. Name the three types of layers found in a Neural Network.
23. Stand for LSTM.
24. Full form of CNN and RNN.

*** 3-Marks Questions**

25. Explain the structure of a Neural Network with its Input, Hidden and Output layers.
26. Explain the process of Forward Propagation and Backpropagation in a Neural Network.
27. Explain how an RNN works and why 'memory' is important for sequential data.
28. Explain Support Vector Machine (SVM), the concept of a hyperplane, support vectors, and the kernel trick.
29. Explain Sigmoid, ReLU and Softmax activation functions with their use cases.

*** 5-Marks Questions**

30. Explain Gradient Boosting and briefly describe how XGBoost and LightGBM improve upon it.
31. Describe Decision Tree and Random Forest algorithms in detail, covering their working, splitting criteria, advantages and weaknesses.
32. Short note :(1) Linear Regression (2) Logistic Regression (3) Naive Bayes
33. Differentiate between CNN and RNN in terms of architecture, working principle and applications
34. Write a program to fruit detect using tensorflow and keras module.

Unit-5: Big Data Implementation

*** 1-Marks Questions**

35. What is Big Data?
36. What is Distributed Computing?
37. What does HDFS stand for?
38. What does RDD stand for?
39. What is Apache Kafka?
40. What does YARN stand for?

*** 3-Marks Questions**

41. Explain the need for Distributed Computing.
42. Explain the architecture of Apache Spark.
43. Explain the main components of the Spark Ecosystem.
44. Explain the features of Kafka.
45. What is RDD? Explain its properties and operations.

*** 5-Marks Questions**

46. What is Distributed Computing? Explain advantage,disadvantage and example.
47. What is Apache Kafka? Explain its architecture and working with a flow.
48. What is Hadoop? Explain its core components in detail.
49. Explain the role of YARN in Hadoop with its components.